American Computer Science League

2025-26 Finals ● Short Problems Solutions ● Junior Division

1. Boolean Algebra

$(A + \overline{B})(\overline{B}\,\overline{C}) + A\overline{B}C$ $= A\,\overline{B}\,\overline{C} + \overline{B}\,\overline{B}\,\overline{C} + A\,\overline{B}\,C$
$= A\,\overline{B}\,\overline{C} + \overline{B}\,\overline{C} + A\,\overline{B}\,C$
$= \overline{B}\,\overline{C}\,(A + 1) + A\,\overline{B}\,C$
$= \overline{B}\,\overline{C} + A\,\overline{B}\,C$

If $B = 1$, then $\overline{B}\,\overline{C} + A\,\overline{B}\,C = 0 * \overline{C} + A * 0 * C = 0$.
If $B = 0$, then $\overline{B}\,\overline{C} + A\,\overline{B}\,C = 1 * \overline{C} + A * 1 * C = \overline{C} + A\,C$
If $\overline{C} = 1$, $\overline{C} + A\,C = 1 + A * 0 = 1$ so (*,0,0) makes it TRUE.
If $\overline{C} = 0$, $\overline{C} + A\,C = 0 + A * 1 = A$ so (1,0,1) makes it TRUE.
Therefore, there are 3 ordered triples that make the expression TRUE.

B. 3

2. Boolean Algebra

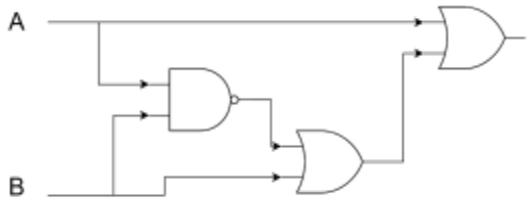
$A(\overline{B} + \overline{C}) + \overline{A}\,\overline{B}(\overline{C + \overline{A}})$ $= A\overline{B} + A\overline{C} + \overline{A}\,\overline{B}(\overline{C}\,A)$
$= A\overline{B} + A\overline{C} + \overline{A}\,\overline{B}\overline{C}\,A$
$= A\overline{B} + A\overline{C} + 0$
$= A\overline{B} + A\overline{C}$

C. $A\overline{B} + A\overline{C}$

3. Digital Electronics

For the digital circuit at the right, the Boolean expression is:

$A + (\overline{A\,B} + B)$
$= A + \overline{A} + \overline{B} + B$
$= 1 + 1 = 1$

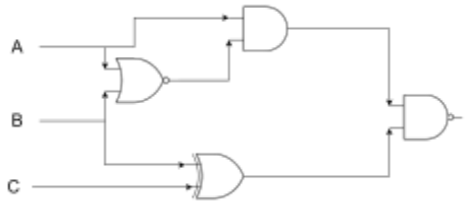


D. 1

4. Digital Electronics

The Boolean expression for the digital circuit below is:

$$\overline{((A(\overline{A+B}))(B \oplus C)}$$
$$= \overline{A(\overline{A+B})} + \overline{B \oplus C}$$
$$= \overline{A} + A + B + \overline{B \oplus C}$$
$$= 1 + B + \overline{B \oplus C} = 1$$



The circuit is always TRUE, so 8 ordered triples make it TRUE.

D. 8

5. Bit-String Flicking

Let X = abcde.

LHS = (NOT 00011) AND (LCIRC-2 (RSHIFT-2 01011)) OR X
= (NOT 00011) AND (LCIRC-2 (RSHIFT-2 01011)) OR abcde
= 11100 AND (LCIRC-2 00010) OR abcde
= 11100 AND 01000 OR abcde
= 01000 OR abcde
= a1cde
RHS = 01101
LHS = RHS → a1cde = 01101 → a=0, b=*, c=1, d=0, e=1 → 0*101

D. 0*101

6. Bit-String Flicking

(RSHIFT-2 $9A_{16}$) OR (RCIRC-2 11000101_2) AND
(LCIRC-2 (NOT 361_8))
= (RSHIFT-2 10011010) OR (RCIRC-2 11000101) AND
(LCIRC-2 (NOT 11110001))
= 00100110 OR 01110001 AND (LCIRC-2 00001110)
= 00100110 OR (01110001 AND 00111000)
= 00100110 OR 00110000
= 00110110

A. 00110110

7. Recursive Functions

$f(21) = f(21-3) + 1 = f(18) + 1 = 42 + 1 = 43$
$f(18) = f(18-3) + 1 = f(15) + 1 = 41 + 1 = 42$
$f(15) = 2 * f(15-1) + 3 = f(14) + 1 = 2 * 19 + 3 = 41$
$f(14) = f(14-3) + 1 = f(11) + 1 = 18 + 1 = 19$
$f(11) = 2 * 11 - 4 = 18$
So $f(21) = 43$

C. 43

8. Recursive Functions

$f(10, 7) = f(7, 10 - 2) - 2 = f(7, 8) - 2 = 4 - 2 = 2$
$f(7, 8) = f(8, 7 - 2) - 2 = f(8, 5) - 2 = 6 - 2 = 4$
$f(8, 5) = f(5, 8 - 2) - 2 = f(5, 6) - 2 = 8 - 2 = 6$
$f(5, 6) = f(6, 5 - 2) - 2 = f(6, 3) - 2 = 10 - 2 = 8$
$f(6, 3) = (6 - 1)(3 - 1) = 5 * 2 = 10$

E. None of the above

9. Prefix/Infix/Postfix

+ + a * b − ^ a 2 c * a b
= + + a * b − (^ a 2) c (* a b)
= + + a * b [− (a 2 ^) c] (a b *)
= + + a [* b (a 2 ^ c −)] (a b *)
= + + a [b a 2 ^ c − *] (a b *)
= + (a b a 2 ^ c − * +) (a b *)
= a b a 2 ^ c − * + a b * +

A. aba2^c−*+ab*+

10. Prefix/Infix/Postfix

4 9 8 & 7 3 # + * 2 6 2 # ^ 3 * 5 1 + 4 & / −
= 4 (9 8 &) (7 3 #) + * 2 (6 2 #) ^ 3 * (5 1 +) 4 & / −
= 4 (9 3 +) * (2 2 ^) 3 * (6 4 &) / −
= (4 12 *) (4 3 *) 6 / −
= 48 (12 6 /) −
= 48 2 −
= 46

B. 46

11. Computer Number System

$EAFBDC_{16}$
$= 1110\ 1010\ 1111\ 1011\ 1101\ 1100_2$
$= 111\ 010\ 101\ 111\ 101\ 111\ 011\ 100_2$
$= 72575734_8$

There are 3 7s, 2 5s, 1 4, 1 3, and 1 2.

D. 7

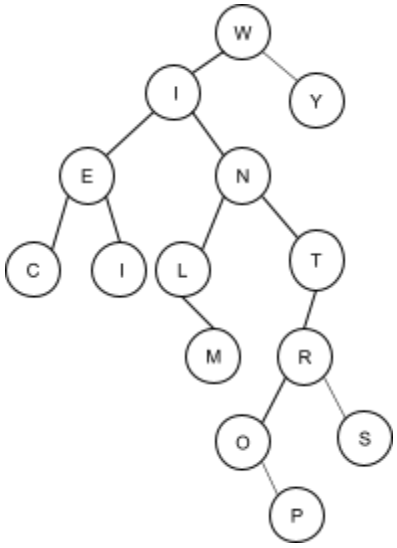
12. Computer Number Systems

$2_{16} + 20_8 + 202_{16} + 2026_8$
$= 10_2 + 10000_2 + 1000000010_2 + 10000010110_2$
$= 11\ 000\ 101\ 010_2$
$= 3\quad 0\quad 5\quad 2_8$

C. 3052

13. Data Structures

The binary search tree for **WINTEROLYMPICS** is:



The depth is 6.

C. 6

14. Data Structures

The following would be popped to eliminate the cheese: B, WBP, P ,L, S, WBP, C - in that order. That's 7 pops.

The following would be pushed to complete the sandwich: WSB, S, L, P, WBP, B - in that order. That's 6 pushes.

A. 7 POP and 6 PUSH

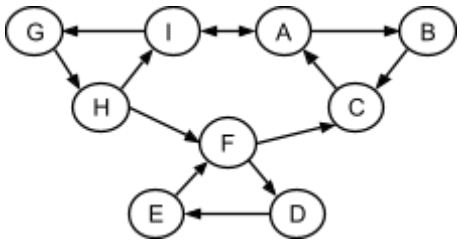
15. Graph Theory

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

ADA, ACB, ACD, ACE, ADF, CDA, CBE, CDF, DAC, DAD so there are 10. ADA and DAD) are not counted since the vertices have to be different. So the answer is 8.

B. 8

16. Graph Theory



The cycles are AIA, FDEF, ABCA, GHIG, AIGHFCA. There are 5 of them.

C. 5

17. What Does This Program Do? (Branching)

```
input num
if num % 2 == 0 then
   if num % 3 == 0 then
      output num
   else
      if num % 4 == 0
         then output num
      end if
   end if
else
   if num % 5 == 0 then
      output num
   end if
end if
```

This outputs even numbers that are divisible by 3, multiples of 4 that are not divisible by 3, and odd numbers that are divisible by 5.

Therefore, the only number not output is 70.

D. 70

18. What Does This Program Do? (Loops)

```
s = 1
sum = 0
for n = 1 to 10000
   term =
   sum = sum + term
   s = -1 * s
next x
pi = sum * 4
output pi
```

The s value swaps between positive and negative fractions so the denominator needs to be odd numbers starting with 1/1. Creating a table helps you come up with an expression. The expression 2*n-1 works for each of them.

n	1	2	3	4	5
den	1	3	5	7	9

A. `s/(2*n-1)`

19. What Does This Program Do? (Arrays)

```
for x = 0 to 4
   for y = x+1 to 5
      if arr(x) > arr(y) then
         t = arr(x)
         arr(x) = arr(y)
         arr(y) = t
      end if
   next y
next x
```

x = 0: 1 5 7 9 3 2
x = 1: 1 2 7 9 5 3
x = 2: 1 2 3 9 7 5,
Therefore, arr(5)=5.

B. 5

20. What Does This Program Do? (Strings)

```
s = "ACSLFINALSCOMPETITION"
ans = ""
for x = 0 to 9
   y = len(s)-x-1
   if s[x] > s[y] then
      ans = ans + s[x]
   else
      ans = ans + s[y]
   end if
next x
output ans
```

This program compares letters from the beginning of the string moving forwards with letters from the end moving backwards.and prints the one that is largest. The pairs of letters are AN, CO, SI, LT, FI, IT, NE, AP, LM, SO. So the answer is "NOSTITNPMS".

B. NOSTITNPMS